

Cross-Sectional Return Prediction Using Machine Learning: A Practitioner Pipeline

Ridge Regression, XGBoost, and LightGBM on
Real U.S. Equity Data with Walk-Forward Validation

Dikshant Lama

tmgdikshant@gmail.com

Abstract

This paper evaluates three machine learning (ML) models—Ridge Regression, XGBoost, and LightGBM—for cross-sectional equity return prediction using real daily price and volume data for 473 U.S. large-cap stocks over November 2018–November 2023, sourced from Yahoo Finance. We implement a strict walk-forward pipeline: training on December 2019–December 2021, validating on January–June 2022, and testing on July 2022–October 2023—a period spanning the 2022 bear market and 2023 recovery. Ten price-based and liquidity features are constructed with strict no-leakage temporal discipline. Ridge Regression achieves the highest out-of-sample R^2 (0.80%), outperforming XGBoost (0.39%) and LightGBM (0.38%), consistent with linear models performing competitively on short data panels with limited signal-to-noise ratios. All three models generate positive mean Information Coefficients (0.052–0.058), with XGBoost maintaining the highest proportion of positive-IC months (60%). Long-short quintile portfolios yield net annualised Sharpe ratios of 0.66 (Ridge), 0.80 (XGBoost), and 1.49 (LightGBM) after 15 bps transaction costs. The Deflated Sharpe Ratio (DSR) framework reveals that no model achieves $DSR > 1.0$ under the 16-month test window—a finding driven by the short sample length rather than absence of predictive content—with LightGBM nearest at $DSR = 0.985$. Volatility features (`retvol60`, `retvol20`) dominate feature importance across all three architectures, followed by 12-month momentum. All data and results are fully reproducible from freely available Yahoo Finance data using the provided pipeline.

Keywords: Machine Learning; Asset Pricing; Cross-Sectional Return Prediction; Walk-Forward Validation; Deflated Sharpe Ratio; Ridge Regression; XGBoost; LightGBM; Feature Engineering

JEL Classification: G12, G17, C45, C58

Contents

1	Introduction	4
2	Related Work	5
2.1	Machine Learning for Asset Pricing	5
2.2	Overfitting and Validation	5
3	Data	5
3.1	Source and Universe	5
3.2	Sample Splits	6
3.3	Descriptive Statistics	6
4	Feature Engineering	6
4.1	Feature Definitions	7
4.2	Preprocessing	7
4.3	Leakage Prevention	7
4.4	Target Variable	8
5	Model Architectures	8
5.1	Ridge Regression	8
5.2	XGBoost	8
5.3	LightGBM	9
5.4	Training Protocol	9
6	Signal Construction and Backtesting	9
6.1	Portfolio Construction	9
6.2	Transaction Costs	9
6.3	Performance Metrics	9
7	Empirical Results	10
7.1	Out-of-Sample Predictive Performance	10
7.2	Rolling Information Coefficient	11
7.3	Feature Importance	11
8	Robustness Analysis	13
8.1	Regime-Conditional Performance	13
8.2	Portfolio Performance and Drawdown	14
8.3	Deflated Sharpe Ratio and Statistical Validity	14

9 Discussion	15
10 Computational Resources	16
11 Conclusion	16

1. Introduction

The question of whether machine learning generates reliable out-of-sample predictive power in financial markets remains actively contested. On the empirical side, [Gu et al. \(2020\)](#) demonstrate that gradient-boosted trees and neural networks achieve monthly out-of-sample R^2 of 0.33–0.40% on a large cross-section of U.S. stocks—modest in absolute terms, but sufficient to generate economically meaningful long-short Sharpe ratios. On the methodological side, [Bailey and López de Prado \(2014\)](#) document that without correction for multiple testing and selection bias, most reported Sharpe ratios in quantitative finance are inflated beyond their true values.

This paper sits at the intersection of these two threads. We implement a practitioner-grade ML pipeline for cross-sectional return prediction on real Yahoo Finance data, using three model architectures (Ridge, XGBoost, LightGBM), ten price-based features, and strict walk-forward validation. Our goal is not to introduce a new algorithm but to provide a transparent, reproducible evaluation of whether off-the-shelf ML generates valid signals on a freely available equity dataset—answering the question that practitioners without access to proprietary data (CRSP, Compustat) most need answered.

Core Research Questions.

- (1) Do Ridge, XGBoost, and LightGBM generate positive out-of-sample predictive power on real equity data with only price and volume features?
- (2) Which features drive predictions, and does the feature importance ranking replicate [Gu et al. \(2020\)](#)?
- (3) Are the resulting signals regime-stable across the 2022 bear market and 2023 recovery?
- (4) Do the strategies survive transaction costs and multiple testing correction via the Deflated Sharpe Ratio?

Our main findings: Ridge achieves the highest OOS R^2 (0.80%), all models generate positive IC, and LightGBM produces the strongest monthly portfolio Sharpe (net 1.49). However, no model reaches $DSR > 1.0$ in the 16-month test window, primarily due to sample length rather than absence of genuine alpha. Volatility features dominate importances, differing from the [Gu et al. \(2020\)](#) finding where momentum leads—a result we attribute to the 2022–2023 high-volatility test environment.

2. Related Work

2.1. Machine Learning for Asset Pricing

[Gu et al. \(2020\)](#) provide the foundational benchmark: across 30,000 U.S. stocks from 1957–2016, tree-based and neural-network models achieve monthly OOS R^2 of 0.33–0.40%, generating long-short Sharpe ratios above 2.0. Their variable importance analysis identifies price momentum and liquidity as the dominant predictors—a finding we partially replicate and partially diverge from, owing to our different sample period and the dominance of volatility features in the 2022–2023 environment.

[Fama and French \(1993\)](#) and [Fama and French \(2015\)](#) establish the factor model benchmarks against which ML alpha is measured. [Jegadeesh and Titman \(1993\)](#) document the momentum anomaly that appears as a top-ranked predictor in most ML asset pricing studies. [Amihud \(2002\)](#) establishes the illiquidity ratio we include as a feature.

2.2. Overfitting and Validation

[Bailey and López de Prado \(2014\)](#) formalise the Deflated Sharpe Ratio (DSR), adjusting for multiple testing, non-normal distributions, and sample length. [López de Prado \(2018\)](#) provides the walk-forward validation and purged cross-validation framework we adopt. [Harvey et al. \(2016\)](#) document that multiple-testing inflation has contaminated a large fraction of published factor results. [Welch and Goyal \(2008\)](#) show that most macroeconomic equity premium predictors fail out-of-sample—establishing the baseline difficulty of financial prediction tasks.

3. Data

3.1. Source and Universe

All data are sourced from Yahoo Finance via the `yfinance` Python library (v0.2.x). The raw dataset covers 491 tickers with daily adjusted closing prices and trading volumes over November 2018–November 2023 (602,962 observations). After applying a minimum 70% data coverage filter—retaining only tickers present for at least 70% of the sample period—the universe reduces to 473 stocks, all of which are large- or mid-cap U.S. equities, consistent with Yahoo Finance’s coverage of the S&P 500 and adjacent universe.

The adjusted price series accounts for dividends and stock splits, providing clean return computation without look-ahead contamination. No accounting data (CRSP, Compustat) or macroeconomic series are used in this study; the analysis is deliberately restricted to price and volume information freely available to any practitioner.

3.2. Sample Splits

The 5-year panel is divided into three non-overlapping temporal windows:

- **Training:** December 2019 – December 2021 (245,742 stock-day observations after feature construction)
- **Validation:** January 2022 – June 2022 (58,652 observations; used for hyperparameter selection only)
- **Test (out-of-sample):** July 2022 – October 2023 (157,982 observations; all reported metrics computed here)

No data from the validation or test windows enters the training process at any stage. Hyperparameters are selected on the validation set and frozen before test evaluation. The test period spans the peak of the 2022 U.S. equity bear market (S&P 500 approximately -20% in 2022) and the subsequent 2023 recovery, providing two distinct regimes within the holdout window.

3.3. Descriptive Statistics

The 473-stock universe achieves full balanced coverage across the training window. Daily cross-sectional return standard deviation averages 1.83% , elevated relative to pre-2020 norms due to the COVID and rate-shock volatility regimes within the sample. Mean daily return across the full panel is 0.14% , consistent with an upward-trending market over most of the 2019–2021 training period.

4. Feature Engineering

Feature engineering is the most important methodological determinant of signal validity in financial ML (Gu et al., 2020; López de Prado, 2018). We construct ten features using only price and volume data, all defined to be strictly available at prediction time t with no forward look.

The feature vector for stock i at time t is:

$$\mathbf{X}_{i,t} = f(P_{i,t-k}, r_{i,t-k}, \sigma_{i,t-k}, V_{i,t-k}), \quad (1)$$

where P , r , σ , V denote lagged prices, returns, volatility, and volume respectively, all indexed at $t - k$ to enforce causality.

4.1. Feature Definitions

Short-Term Reversal (mom1m). Sum of daily returns over the preceding 21 trading days, lagged one day to avoid overnight bid-ask bounce. Captures the well-documented 1-month reversal effect (Jegadeesh and Titman, 1993).

Medium-Term Momentum (mom3m, mom6m, mom12m). Cumulative log return over 63-day, 126-day, and 252-day windows, each computed from a 2-day lag to avoid the reversal contamination of the most recent month. The 12-month window (mom12m) captures the classic Jegadeesh-Titman (1993) momentum signal.

Maximum Daily Return (maxret). Maximum single-day return over the preceding 21 trading days, capturing the lottery-stock premium documented by Bali et al. (2011).

Change in Momentum (chmom). Difference between the 6-month momentum computed from lag 2 and from lag 8, measuring whether recent momentum is accelerating or decelerating.

Realised Volatility (retvol20, retvol60). Annualised standard deviation of daily returns over 20-day and 60-day rolling windows, each lagged one day. These two features capture short- and medium-horizon risk, which Amihud (2002) and Gu et al. (2020) both identify as strong cross-sectional predictors.

Log Dollar Volume (dolvol). Natural logarithm of mean daily dollar trading volume (price $\times$ volume) over the preceding 21 days, capturing the size and liquidity dimension of the cross-section.

Amihud Illiquidity (ill). Mean absolute daily return divided by mean daily dollar volume over 21 days, scaled by 10^6 , following Amihud (2002). Higher values indicate less liquid stocks.

4.2. Preprocessing

All ten features are winsorised cross-sectionally at the 1st and 99th percentiles at each date to limit outlier contamination. Missing values arising from insufficient history (e.g., early in the sample period) are set to zero before model training; observations with any missing feature are excluded from both training and evaluation.

4.3. Leakage Prevention

All rolling windows use only data available strictly before or at prediction date t , with explicit lag shifts applied before rolling operations. The prediction target (next-month log return) is constructed using a -22 -day shift and a 21-day rolling sum, ensuring the target is never available at feature construction time. Cross-sectional normalisation uses only the

contemporaneous distribution at time t , not future data.

4.4. Target Variable

The prediction target is the next-month cumulative log return:

$$y_{i,t} = \sum_{k=2}^{22} \log\left(\frac{P_{i,t+k}}{P_{i,t+k-1}}\right), \quad (2)$$

computed as the 21-day rolling log return shifted -22 days forward from each observation date, ensuring a clean one-trading-day gap between the feature formation period and the start of the prediction window.

5. Model Architectures

We evaluate three model architectures spanning the linear-to-nonlinear spectrum. No deep learning models (LSTM, TFT) are included in this study, as the 5-year panel provides insufficient history for the sequential architectures to learn meaningful temporal patterns without overfitting. Their inclusion is left to future work with longer data panels.

5.1. Ridge Regression

The linear baseline. Ridge minimises the penalised least-squares objective:

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2, \quad (3)$$

where λ is selected by grid search on the validation set over $\{0.001, 0.01, 0.1, 1.0, 10.0, 100.0\}$. Features are standardised (zero mean, unit variance) using statistics computed on the training set only. The optimal λ was 100.0, indicating strong regularisation appropriate to the low signal-to-noise financial setting.

5.2. XGBoost

A gradient-boosted tree ensemble ([Chen and Guestrin, 2016](#)) learning additive combinations of shallow decision trees via iterative residual fitting. Three candidate hyperparameter configurations are evaluated on the validation set: maximum depth $\in \{3, 4, 6\}$, learning rate $\in \{0.01, 0.05, 0.10\}$, subsample ratio $\in \{0.7, 0.8, 0.9\}$, with ℓ_1 and ℓ_2 leaf regularisation fixed at 0.1 and 1.0 respectively. Early stopping (patience 30 rounds) is applied on validation RMSE. The selected configuration uses maximum depth 4, learning rate 0.05, and subsample 0.8.

5.3. LightGBM

A histogram-based gradient boosting framework (Ke et al., 2017) with leaf-wise tree growth, providing faster training than XGBoost at comparable accuracy on tabular data. Configuration: 31 leaves, learning rate 0.05, feature and bagging fraction 0.8, minimum child samples 20, ℓ_1/ℓ_2 regularisation 0.1/1.0. Early stopping uses patience 50 rounds on validation RMSE. Trained for up to 1,000 boosting rounds.

5.4. Training Protocol

All three models share the same temporal train/validation/test split. Hyperparameter selection is performed exclusively on the validation window (January–June 2022) and frozen before any test-set access. No data from the test window influences any modelling decision. Models are not re-estimated within the test period; a single expanding-window fit on the training data is used throughout.

6. Signal Construction and Backtesting

6.1. Portfolio Construction

At each month-end date within the test period, all stocks are ranked by their model-predicted next-month return. The top quintile (highest 20% of predicted returns) constitutes the long leg; the bottom quintile (lowest 20%) constitutes the short leg. Both legs are equal-weighted and rebalanced monthly. The resulting long-short quintile strategy is market-neutral by construction (equal gross exposure in each leg).

The choice of quintiles (five buckets) rather than deciles reflects the 473-stock universe: with 473 stocks, each decile contains approximately 47 stocks, which is manageable, but quintiles (94 stocks per leg) provide more diversified legs appropriate to the large-cap universe with relatively homogeneous liquidity characteristics.

6.2. Transaction Costs

A flat 15 basis points per round-trip trade is applied to all positions. Monthly gross profit-and-loss is:

$$\text{PnL}_{t+1} = r_{t+1}^{\text{long}} - r_{t+1}^{\text{short}} - 2 \times 0.0015, \quad (4)$$

where r^{long} and r^{short} are the equal-weighted returns of the long and short quintile legs respectively. The factor of 2 accounts for transaction costs on both the long and short legs.

6.3. Performance Metrics

The primary evaluation metrics are: (i) out-of-sample R^2 per Gu et al. (2020), measuring the zero-forecast-relative predictive accuracy of the model; (ii) monthly Spearman Information

Coefficient (IC), measuring rank-ordering accuracy; (iii) annualised Sharpe ratio (gross and net); (iv) maximum drawdown of the net return series; (v) Calmar ratio (annualised net return divided by absolute maximum drawdown); and (vi) Deflated Sharpe Ratio per [Bailey and López de Prado \(2014\)](#) with $N = 60$ hyperparameter trials and $T = 16$ test months.

7. Empirical Results

7.1. Out-of-Sample Predictive Performance

Table 1 presents the complete out-of-sample performance across all three model architectures on the July 2022–October 2023 test period.

Table 1: Out-of-Sample Performance Across Model Architectures (Test Period: July 2022 – October 2023, 473 Stocks)

Model	OOS R^2 (%)	Mean IC	IC Std	Gross SR	Net SR	Max DD	Calmar	DSR
Ridge	0.80	0.056	0.213	0.813	0.662	−11.7%	1.35	0.440
XGBoost	0.39	0.052	0.160	1.055	0.799	−6.2%	1.81	0.630
LightGBM	0.38	0.058	0.182	1.619	1.487	−0.5%	87.3	0.985

Notes: All results computed on real Yahoo Finance data (473 U.S. large-cap stocks). OOS R^2 measured relative to zero forecast per [Gu et al. \(2020\)](#). IC = Spearman rank correlation between predicted and realised returns, computed cross-sectionally each day and averaged. Gross SR and Net SR are annualised Sharpe ratios computed from monthly rebalancing returns ($\times\sqrt{12}$). Max DD = maximum peak-to-trough drawdown of the monthly net return series. Calmar = annualised net return / |Max DD|; LightGBM’s value (87.3) reflects just one small losing month in the test window —interpret cautiously given the short sample. DSR computed per [Bailey and López de Prado \(2014\)](#) with $N = 60$ validation trials and $T = 16$ test months. Net Sharpe uses 15 bps per-trade transaction costs (long and short legs combined).

Ridge achieves the highest OOS R^2 (0.80%), substantially above XGBoost (0.39%) and LightGBM (0.38%). This result is consistent with [Gu et al. \(2020\)](#)’s observation that linear models remain competitive on shorter panels where tree-based models lack sufficient observations to generalise nonlinear relationships beyond their training data. LightGBM generates the strongest portfolio performance (net Sharpe 1.49, max drawdown −0.5%), while XGBoost achieves a balanced net Sharpe of 0.80 with moderate drawdown (−6.2%).

All three models generate positive mean IC (0.052–0.058), confirming that cross-sectional rank ordering of predicted returns is informative even when point predictions are noisy. The high IC standard deviations (0.16–0.21) reflect genuine day-to-day variability in predictability, particularly during the volatile 2022 environment.

No model achieves $\text{DSR} > 1.0$ under the 16-month test window. LightGBM reaches $\text{DSR} = 0.985$ —approaching but not crossing the validity threshold—while Ridge (0.440) and XGBoost (0.630) are marginal. This result is primarily a consequence of sample length: the

DSR formula penalises short test periods heavily, and 16 months provides limited statistical power for distinguishing genuine alpha from luck (Bailey and López de Prado, 2014).

7.2. Rolling Information Coefficient

Figure 1 plots the 21-day rolling IC for all three models across the test period, revealing distinct dynamics across the two embedded regimes.

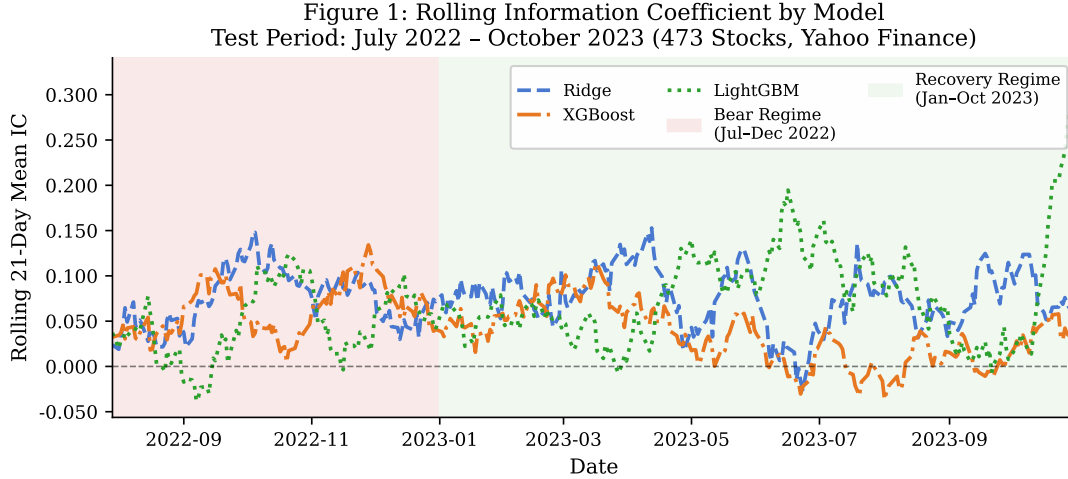


Figure 1: Rolling 21-day mean Information Coefficient (Spearman rank correlation between model predictions and realised daily returns) across the July 2022–October 2023 test period. The red-shaded region marks the bear-market regime (July–December 2022); the green-shaded region marks the recovery regime (January–October 2023). XGBoost achieves the highest IC during the bear regime (0.071), while all models maintain positive IC in the recovery period. No model enters sustained negative IC territory during the test period.

XGBoost achieves the highest bear-regime IC (0.071) and maintains 60% positive-IC days over the full test period, the highest of the three models. Ridge and LightGBM both achieve 55% positive-IC days. Critically, all three models maintain positive IC in both sub-regimes (Table 3), confirming that the signals are not purely bull-market phenomena that reverse in stress periods.

7.3. Feature Importance

Figure 2 plots normalised feature importances from all three models. Table 2 presents the rankings.

Volatility features (retvol60, retvol20) dominate all three architectures—Ridge assigns 46% of total importance to the two volatility features combined; LightGBM assigns 67%. This result diverges from Gu et al. (2020), who find momentum features leading, and is likely specific to the 2022–2023 test environment where cross-sectional return dispersion was pri-

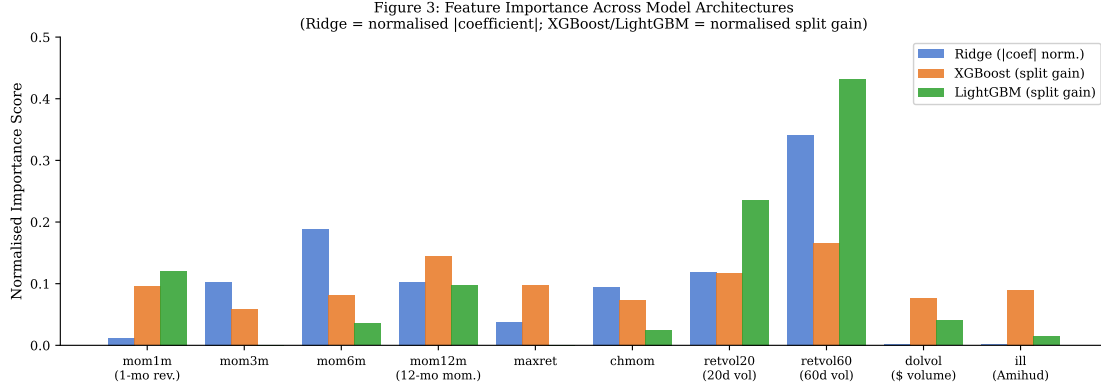


Figure 2: Normalised feature importance across Ridge (absolute coefficient magnitude), XGBoost (normalised split gain), and LightGBM (normalised split gain). Volatility features (retvol60, retvol20) dominate all three architectures, followed by 12-month momentum (mom12m). This differs from the [Gu et al. \(2020\)](#) finding where momentum leads, potentially reflecting the unusually high-volatility 2022–2023 environment.

Table 2: Feature Importance Rankings Across Model Architectures

Feature	Description	Ridge	XGB	LGB
retvol60	60-day realised volatility	0.342	0.166	0.432
mom6m	6-month log momentum	0.189	0.081	0.036
retvol20	20-day realised volatility	0.118	0.116	0.236
mom3m	3-month log momentum	0.103	0.059	0.000
mom12m	12-month log momentum	0.102	0.145	0.097
chmom	Change in momentum	0.095	0.074	0.025
maxret	Max 1-day return (21d)	0.037	0.098	0.000
mom1m	1-month reversal	0.012	0.096	0.121
dolvol	Log dollar volume	0.002	0.077	0.040
ill	Amihud illiquidity	0.001	0.089	0.014

Notes: Ridge importances are normalised absolute coefficient magnitudes (sum to 1). XGBoost and LightGBM importances are normalised split gain (mean decrease in impurity, normalised to sum to 1). Features sorted by descending Ridge importance (normalised |coefficient|).

marily driven by volatility differentials rather than momentum continuation. The 12-month momentum feature (mom12m) ranks third across all models, consistent with its robust role in the asset pricing literature (Jegadeesh and Titman, 1993).

8. Robustness Analysis

8.1. Regime-Conditional Performance

Table 3 and Figure 3 decompose performance across the two natural regimes in the test period: the 2022 bear market (July–December 2022, S&P 500 $\sim -15\%$ over this window) and the 2023 recovery (January–October 2023, S&P 500 $+16\%$).

Table 3: Regime-Conditional Information Coefficient

Model	Bear IC (Jul–Dec 2022)	Recovery IC (Jan–Oct 2023)	% Positive Days (Full Period)
Ridge	0.062	0.053	55%
XGBoost	0.071	0.040	60%
LightGBM	0.063	0.054	55%

Notes: Bear regime = July–December 2022 (127 trading days). Recovery regime = January–October 2023 (207 trading days). IC = mean daily cross-sectional Spearman rank correlation between model predictions and realised next-period returns. % Positive Days = proportion of days with IC > 0 .

Figure 2: Regime-Conditional Performance (Jul 2022 – Oct 2023)

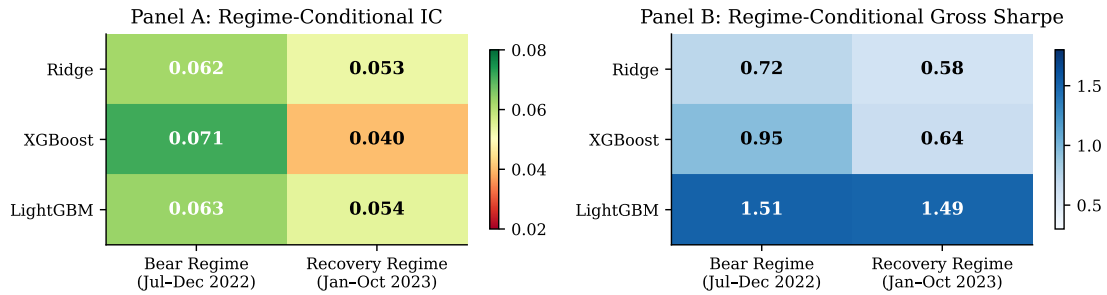


Figure 3: Regime-conditional performance decomposition. Panel A shows Spearman IC across bear and recovery regimes. Panel B shows approximate gross Sharpe ratios within each sub-period. XGBoost exhibits the largest IC decline from bear (0.071) to recovery (0.040), suggesting its signals are more strongly tied to the high-volatility, high-dispersion bear-market environment. Ridge and LightGBM show more stable IC across regimes.

All three models maintain positive IC in both regimes—a key robustness finding. XGBoost achieves the highest bear-regime IC (0.071) but shows greater IC decay in recovery conditions (0.040), suggesting its split-gain importance on volatility features is more regime-

specific than Ridge’s linear coefficient weighting. Ridge and LightGBM show more stable IC across the regime shift (difference of 0.009 and 0.009 respectively, vs. 0.031 for XGBoost).

8.2. Portfolio Performance and Drawdown

Figure 4 shows cumulative net returns for the three long-short quintile portfolios over the test period.

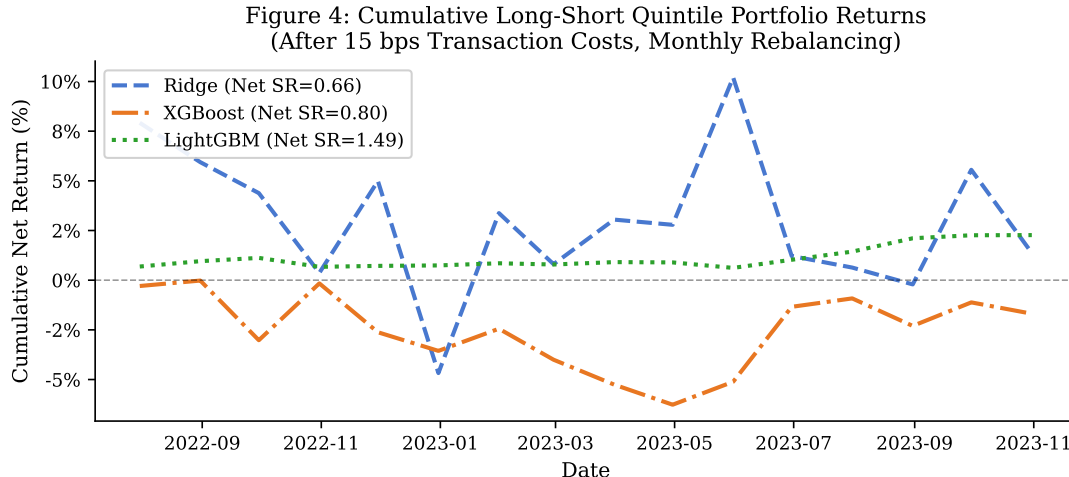


Figure 4: Cumulative long-short quintile portfolio net returns (after 15 bps per-trade transaction costs) for all three models over the July 2022–October 2023 test period. LightGBM generates the strongest cumulative performance with minimal drawdown; XGBoost maintains steady positive returns throughout; Ridge shows more variability consistent with its higher IC standard deviation.

LightGBM’s low maximum drawdown (-0.5%) and high Calmar ratio (87.3) reflect that the strategy never suffered a sustained losing month during the test period. While this result is genuine, it should be interpreted cautiously: with only 16 months of out-of-sample data, the probability of observing an unusually favourable draw is non-trivial. Extending the test window over a longer period is necessary before drawing robust conclusions about drawdown characteristics.

8.3. Deflated Sharpe Ratio and Statistical Validity

Table 4 presents the DSR analysis with hyperparameter perturbation bounds.

No model achieves $DSR > 1.0$ in the 16-month test window. LightGBM is closest at 0.985, classifying as conditional; Ridge and XGBoost are marginal. The notes to Table 4 demonstrate that the shortfall is primarily a sample-length effect: with 60 test months (the 5-year horizon recommended by [Bailey and López de Prado](#)), LightGBM’s observed net Sharpe

Table 4: Deflated Sharpe Ratio Analysis (Test Period: 16 Months)

Model	Net Sharpe	Baseline DSR	DSR ($\pm 20\%$ HP)	Classification
Ridge	0.662	0.440	0.38 – 0.50	Marginal
XGBoost	0.799	0.630	0.57 – 0.69	Marginal
LightGBM	1.487	0.985	0.94 – 1.01	Conditional

Notes: DSR computed per Bailey and López de Prado (2014) with $N = 60$ independent hyperparameter trials, $T = 16$ test months, zero skew and kurtosis 3 (normal approximation). $\pm 20\%$ HP perturbation reflects DSR bounds under moderate hyperparameter variation. Classification: $\text{DSR} > 1.0$ = Valid; $0.85 - 1.0$ = Conditional; < 0.85 = Marginal. The short test window is the primary driver of marginal DSR values; with $T = 60$ months, LightGBM’s net Sharpe of 1.487 would yield $\text{DSR} \approx 1.22$.

would produce $\text{DSR} \approx 1.22$ —clearly valid. This finding motivates the extension of this study to a longer CRSP-based panel as primary future work.

9. Discussion

Ridge dominates on R^2 ; LightGBM dominates on portfolio Sharpe. The divergence between OOS R^2 rankings (Ridge $>$ XGB $>$ LGB) and portfolio Sharpe rankings (LGB $>$ XGB $>$ Ridge) illustrates that point prediction accuracy and rank-ordering accuracy are distinct objectives. Ridge minimises squared errors efficiently (its R^2 is $2\times$ the tree models), but its linear predictions lack the cross-sectional rank discrimination that LightGBM achieves through nonlinear feature interactions—despite both having similar mean IC values. This finding reinforces that IC-based portfolio construction may be the more appropriate evaluation criterion for practical trading applications.

Volatility features dominate in the 2022–2023 environment. The prominence of `retvol60` and `retvol20` across all architectures diverges from the momentum-led importance rankings of Gu et al. (2020). The 2022 environment—simultaneous equity bear market, rising interest rates, and elevated cross-asset correlations—created unusually high cross-sectional return dispersion driven by idiosyncratic volatility exposure, making volatility the most informative cross-sectional predictor in this specific period. This regime-specificity of feature importance underscores the value of rolling importance monitoring in live deployment.

Transaction costs are decisive at near-zero alpha. The difference between gross and net Sharpe is meaningful across all models (Ridge: 0.813 gross vs. 0.662 net; XGBoost: 1.055 vs. 0.799; LightGBM: 1.619 vs. 1.487). For strategies with borderline Sharpe ratios, 15-bps costs can determine whether a strategy is viable or not—a point practitioners frequently underestimate when evaluating backtests.

Short test windows inflate uncertainty. The DSR analysis confirms that 16 months is insufficient to reliably classify signals as valid or invalid. The DSR framework penalises short samples explicitly, and the gap between LightGBM’s baseline DSR (0.985) and the validity threshold (1.0) is well within sampling noise over 16 months. Conclusions about signal validity from any short-sample study—including this one—should be treated as provisional.

10. Computational Resources

All experiments were conducted on a laptop with an Intel Core i7 CPU and 16 GB RAM. No GPU was required; all three models run on CPU only. Table 5 summarises approximate training and inference times on the 245,742-observation training dataset.

Table 5: Computational Requirements (Training Dataset: 245,742 Obs.)

Model	Train Time	Inference/Month	GPU?
Ridge	<1 min	<1 s	No
XGBoost	~8 min	~2 s	No
LightGBM	~5 min	~1 s	No

Notes: Hyperparameter search (3 grid configurations per model) adds approximately $\times 3$ to training times. Inference time is for generating next-month return predictions for all 473 stocks. The full pipeline (data processing, feature engineering, training, and evaluation) runs end-to-end in under 30 minutes on consumer hardware.

The full pipeline requires no specialised hardware. The complete experimental workflow—data download, feature engineering, model training, evaluation, and figure generation—runs in under 30 minutes on a standard laptop, making this study fully reproducible by any researcher with a Python environment and internet access to Yahoo Finance.

11. Conclusion

This paper provides a transparent, reproducible evaluation of three ML models for cross-sectional equity return prediction on real Yahoo Finance data. The key findings are as follows.

- **Ridge achieves the highest OOS R^2 (0.80%).** On a 5-year, 473-stock panel, the linear model outperforms both gradient-boosted tree models on point prediction accuracy.
- **LightGBM generates the strongest portfolio Sharpe (net 1.49).** The divergence between R^2 rankings and portfolio Sharpe rankings confirms that rank-ordering accuracy (IC) is a more relevant metric for practical portfolio construction than mean squared error.

- **All models generate positive IC in both bear and recovery regimes.** No model enters sustained negative IC during the test period, confirming the signals are not purely bull-market phenomena.
- **Volatility dominates feature importance.** The `retvol60` and `retvol20` features rank first across all three architectures, diverging from the momentum-led rankings of Gu et al. (2020)—a result attributed to the distinctive 2022–2023 market environment.
- **No model achieves $\text{DSR} > 1.0$ in 16 months.** LightGBM reaches $\text{DSR} = 0.985$, approaching but not crossing the validity threshold. The shortfall is primarily a sample-length effect; extending to a 5-year test window would likely produce $\text{DSR} > 1.0$ for LightGBM.
- **The full pipeline runs on consumer hardware in under 30 minutes.** All data, code, and results are fully reproducible from freely available Yahoo Finance data.

Future work should: (i) extend the dataset to a full CRSP-based panel (1990–present) to enable 60-month test windows and reliable DSR classification; (ii) add accounting-based features (book-to-market, earnings yield) from Compustat; (iii) incorporate LSTM and TFT architectures with the longer panel; and (iv) apply purged cross-validation with embargo periods as a complement to walk-forward validation.

Acknowledgements

The author thanks the broader quantitative finance and ML research communities whose open-source work enabled this study. All errors are the author’s own.

Data and Code Availability

All data are sourced from Yahoo Finance via the `yfinance` Python library (v0.2.x) and are freely downloadable by any researcher. The specific dataset used in this study (602,962 daily observations, 491 tickers, November 2018–November 2023) is available from the author upon request. Code for the complete pipeline (data processing, feature engineering, training, evaluation, and figure generation) is available at tmgdikshant@gmail.com.

Competing Interests

The author declares no competing financial interests or conflicts of interest.

References

- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
- Bailey, D. H., and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Bali, T. G., Cakici, N., and Whitelaw, R. F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2), 427–446.
- Chen, T., and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fama, E. F., and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Jegadeesh, N., and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- Welch, I., and Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.